

**BELLSHILL AGCM, AB, Canada**WMO: **710100**Lat: **52.5822N**Lon: **111.4653W**Elev: **2365**StdP: **13.48**Time Zone: **-7.00 (NAM)**Period: **07-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -22.6      | -17.7 | -28.5                         | 1.2 | -22.3 | -23.6 | 1.6 | -17.5 | 28.2                  | 20.2 | 25.5 | 21.0 | 7.5                   | 270  | 0.706 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.0                   | 84.7            | 63.1 | 81.1 | 61.4 | 77.9 | 60.4 | 66.5                | 78.9 | 64.5 | 76.8 | 62.7 | 73.6 | 10.8                 | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.1                            | 91.0 | 72.1 | 60.1 | 84.6 | 69.8 | 58.2 | 79.1 | 67.8 | 32.5          | 79.0 | 30.9 | 76.5 | 29.5 | 73.8 | 73.2              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |     |
| (4) 27.1          | 23.5 | 20.6 | DB                         | -29.3 | 90.3               | 4.6 | 4.1                                                | -32.6 | 93.2       | -35.3 | 95.6       | -37.9 | 97.9       | -41.3 | 100.9 | (4) |
| (5)               |      |      | WB                         | -29.5 | 70.4               | 4.5 | 2.3                                                | -32.8 | 72.0       | -35.5 | 73.3       | -38.0 | 74.6       | -41.3 | 76.3  | (5) |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 37.9   | 13.3  | 11.8  | 24.0  | 38.3  | 52.0  | 58.9  | 63.7  | 61.8  | 53.1  | 40.3  | 24.4  | 11.4  |  |  |
|                                                                        | DBStd       | 22.14  | 15.36 | 14.59 | 13.81 | 9.63  | 8.15  | 5.90  | 5.51  | 6.00  | 8.50  | 9.23  | 12.88 | 14.11 |  |  |
|                                                                        | HDD50       | 5807   | 1139  | 1068  | 805   | 364   | 75    | 7     | 0     | 1     | 63    | 319   | 770   | 1196  |  |  |
|                                                                        | HDD65       | 10000  | 1604  | 1488  | 1270  | 800   | 407   | 197   | 91    | 133   | 366   | 764   | 1219  | 1661  |  |  |
|                                                                        | CDD50       | 1393   | 0     | 0     | 0     | 14    | 136   | 275   | 425   | 367   | 155   | 20    | 1     | 0     |  |  |
|                                                                        | CDD65       | 113    | 0     | 0     | 0     | 0     | 4     | 15    | 52    | 34    | 8     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 1706   | 0     | 0     | 0     | 6     | 131   | 187   | 658   | 517   | 201   | 6     | 0     | 0     |  |  |
|                                                                        | CDH80       | 420    | 0     | 0     | 0     | 0     | 17    | 28    | 194   | 128   | 53    | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 9.7    | 10.1  | 9.9   | 10.6  | 11.5  | 11.1  | 9.6   | 8.1   | 8.0   | 8.8   | 9.6   | 9.8   | 9.6   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 16.1   | 0.8   | 0.5   | 0.7   | 1.2   | 1.7   | 2.9   | 2.8   | 2.2   | 1.3   | 0.7   | 0.7   | 0.7   |  |  |
|                                                                        | PrecMax     | 21.5   | 2.4   | 0.9   | 1.7   | 2.8   | 4.1   | 4.3   | 5.9   | 4.9   | 3.3   | 1.9   | 2.4   | 1.2   |  |  |
|                                                                        | PrecMin     | 10.3   | 0.2   | 0.1   | 0.0   | 0.2   | 0.4   | 0.7   | 0.7   | 0.2   | 0.0   | 0.2   | 0.1   | 0.1   |  |  |
|                                                                        | PrecStd     | 3.3    | 0.5   | 0.3   | 0.4   | 0.6   | 0.9   | 1.0   | 1.2   | 0.9   | 0.9   | 0.4   | 0.5   | 0.3   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 43.5  | 44.4  | 56.9  | 71.7  | 82.5  | 83.2  | 90.1  | 87.5  | 86.2  | 73.4  | 57.1  | 44.3  |  |  |
|                                                                        |             | MCWB   | 36.9  | 38.5  | 45.2  | 49.4  | 56.0  | 61.9  | 65.1  | 63.1  | 60.8  | 54.3  | 43.7  | 37.9  |  |  |
|                                                                        | 2%          | DB     | 40.3  | 38.5  | 50.6  | 65.8  | 77.8  | 79.1  | 85.4  | 83.4  | 79.9  | 64.7  | 49.1  | 38.6  |  |  |
|                                                                        |             | MCWB   | 35.5  | 34.1  | 41.3  | 47.5  | 54.4  | 60.2  | 65.6  | 62.8  | 57.9  | 50.0  | 40.4  | 33.7  |  |  |
|                                                                        | 5%          | DB     | 37.2  | 35.4  | 44.3  | 59.6  | 73.2  | 75.4  | 81.0  | 79.9  | 74.5  | 59.2  | 43.5  | 34.5  |  |  |
|                                                                        |             | MCWB   | 33.6  | 32.1  | 37.9  | 44.2  | 53.2  | 59.1  | 64.1  | 61.9  | 56.2  | 47.2  | 37.0  | 30.3  |  |  |
|                                                                        | 10%         | DB     | 33.8  | 32.4  | 39.9  | 54.3  | 68.5  | 71.8  | 77.0  | 76.0  | 69.2  | 54.1  | 39.5  | 30.3  |  |  |
|                                                                        |             | MCWB   | 30.6  | 29.9  | 35.1  | 41.9  | 50.9  | 57.6  | 62.8  | 60.9  | 54.0  | 44.0  | 34.5  | 27.5  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 37.8  | 38.6  | 46.0  | 51.2  | 58.2  | 65.5  | 70.2  | 68.2  | 62.3  | 55.3  | 45.7  | 38.4  |  |  |
|                                                                        |             | MCDB   | 42.8  | 43.8  | 55.6  | 70.4  | 77.5  | 77.7  | 82.0  | 79.3  | 79.1  | 71.9  | 55.6  | 43.4  |  |  |
|                                                                        | 2%          | WB     | 35.7  | 34.5  | 41.2  | 47.9  | 55.9  | 63.0  | 67.4  | 65.6  | 59.2  | 51.7  | 41.2  | 34.0  |  |  |
|                                                                        |             | MCDB   | 39.9  | 37.7  | 48.9  | 63.9  | 72.5  | 74.0  | 79.8  | 77.5  | 76.4  | 61.2  | 47.8  | 38.3  |  |  |
|                                                                        | 5%          | WB     | 33.7  | 32.4  | 38.0  | 45.1  | 54.0  | 60.6  | 65.5  | 63.6  | 56.9  | 48.2  | 37.8  | 30.5  |  |  |
|                                                                        |             | MCDB   | 37.1  | 35.2  | 43.8  | 57.6  | 70.4  | 71.2  | 78.0  | 75.9  | 71.5  | 57.5  | 42.8  | 34.1  |  |  |
|                                                                        | 10%         | WB     | 30.7  | 29.7  | 35.4  | 42.6  | 52.1  | 58.8  | 63.8  | 61.7  | 55.0  | 45.2  | 35.1  | 27.5  |  |  |
|                                                                        |             | MCDB   | 33.8  | 32.5  | 39.8  | 53.2  | 66.3  | 69.2  | 74.9  | 73.2  | 68.2  | 53.7  | 38.7  | 30.3  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 16.3  | 16.4  | 16.3  | 19.6  | 22.9  | 20.3  | 22.0  | 22.5  | 22.9  | 19.2  | 15.5  | 14.7  |  |  |
|                                                                        |             | MCDBR  | 16.4  | 18.1  | 21.2  | 29.3  | 29.4  | 25.3  | 28.7  | 29.5  | 32.0  | 27.8  | 19.9  | 18.2  |  |  |
|                                                                        |             | MCWBR  | 13.2  | 14.8  | 13.7  | 14.5  | 12.3  | 11.2  | 12.6  | 12.8  | 13.7  | 15.0  | 12.8  | 14.3  |  |  |
|                                                                        | 5% WB       | MCDBR  | 17.0  | 17.8  | 19.9  | 26.8  | 26.1  | 21.4  | 25.5  | 25.6  | 28.7  | 26.5  | 18.8  | 17.5  |  |  |
|                                                                        |             | MCWBR  | 14.0  | 14.7  | 13.1  | 13.9  | 11.8  | 10.7  | 12.6  | 12.6  | 13.3  | 14.5  | 12.7  | 14.2  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.263 | 0.298 | 0.292 | 0.334 | 0.354 | 0.368 | 0.361 | 0.376 | 0.318 | 0.282 | 0.258 | 0.243 |  |  |
|                                                                        | taud        |        | 2.296 | 2.239 | 2.420 | 2.352 | 2.342 | 2.340 | 2.394 | 2.329 | 2.457 | 2.538 | 2.457 | 2.341 |  |  |
|                                                                        | Ebn at Noon |        | 235   | 256   | 288   | 286   | 284   | 280   | 280   | 269   | 275   | 267   | 241   | 225   |  |  |
|                                                                        | Edh at Noon |        | 19    | 28    | 29    | 35    | 37    | 37    | 35    | 36    | 27    | 20    | 16    | 15    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 347   | 681   | 1145  | 1506  | 1835  | 1892  | 2000  | 1625  | 1147  | 685   | 347   | 259   |  |  |
|                                                                        | RadStd      |        | 30    | 51    | 86    | 143   | 124   | 132   | 99    | 132   | 120   | 92    | 35    | 23    |  |  |

**Historical Trends**

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (75 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page